CHAPTER 15

Quantitative data analysis: Hypothesis testing

LEARNING OBJECTIVES

After completing Chapter 15, you should be able to:

1. Discuss type I errors, type II errors, and statistical power.
2. Test hypotheses using the appropriate statistical technique.
3. Describe data warehousing, data mining, and operations research.
4. Describe useful software packages for quantitative data analysis.

INTRODUCTION

In Chapter 5 we discussed the steps to be followed in hypothesis development and testing. These steps are:

1. State the null and the alternate hypotheses.
2. Determine the level of significance desired ($p = 0.05$, or more, or less).
3. Choose the appropriate statistical test depending on the type of scales that have been used (nominal, ordinal, interval, or ratio).
4. See if the output results from computer analysis indicate that the significance level is met. When the resultant value is larger than the critical value, the null hypothesis is rejected, and the alternate accepted. If the calculated value is less than the critical value, the null hypothesis is accepted and the alternate hypothesis rejected.

In this chapter we will discuss hypothesis testing. First, we will pay attention to type I errors, type II errors, and statistical power. Then, we will discuss various univariate and bivariate statistical tests that can be used to test hypotheses. Finally, we will come back to the Excelsior Enterprises case and test the hypotheses that were developed in the previous chapter.